United States Patent	12397732
Kind Code	B2
Date of Patent	August 26, 2025
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Illumination control device, biometric information acquiring device, and illumination control method

Abstract

An illumination control device includes: an accident detection unit to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; a lamp body control unit to command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected by the accident detection unit; and a lamp color changing unit to change a lamp color of the lamp body when the accident is detected by the accident detection unit.

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Appl. No.:	**18/292614**
Filed (or PCT Filed):	**September 24, 2021**
PCT No.:	**PCT/JP2021/034932**
PCT Pub. No.:	**WO2023/047505**
PCT Pub. Date:	March 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240336215 A1	Oct. 10, 2024

Publication Classification

Int. Cl.: **B60R21/01** (20060101); **B60Q3/70** (20170101); **B60R21/015** (20060101); **G06V40/10** (20220101)

U.S. Cl.:

CPC **B60R21/01538** (20141001); **B60Q3/70** (20170201); **G06V40/103** (20220101); **G06V40/15** (20220101); B60Q2500/30 (20220501); B60R2021/01006 (20130101); B60R2021/01013 (20130101)

Field of Classification Search

CPC: B60R (21/01538); B60R (2021/01006); B60R (2021/01013); B60Q (3/70); B60Q (2500/30); B60Q (3/76); B60Q (3/80); G06V (40/103); G06V (40/15); G06V (20/59); A61B (5/02); A61B (5/08); A61B (5/0077); Y02B (20/40)

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to an illumination control technique of controlling illumination to a vehicle interior.
BACKGROUND ART
(2) Conventionally, there is a technique of lighting an illumination device when imaging a vehicle interior.
(3) For example, Patent Literature 1 discloses a technique of lighting an illumination device such as an indoor light or an imaging light attached to a camera in advance depending on brightness and darkness in a vehicle interior in an emergency notification device that images an occupant when a vehicle accident occurs and transmits an imaged image to an emergency center. In the conventional technique described in Patent Literature 1, imaging can be performed even in a dark case by lighting the illumination device when a vehicle accident occurs. The imaged image is used to visually recognize a situation of the occupant in the emergency center.
CITATION LIST
Patent Literature
(4) Patent Literature 1: JP 2003-306106 A
SUMMARY OF INVENTION
Technical Problem
(5) In the conventional technique described in Patent Literature 1, since the illumination device is simply lighted and illumination is performed using light of a certain color, there is a problem that there is biometric information that cannot be obtained from an imaged image under some imaging conditions when a vehicle accident occurs.
(6) The present disclosure has been made in order to solve the above problem, and an object of the present disclosure is to provide an illumination control technique of controlling illumination in such a manner that biometric information can be obtained without depending on imaging conditions when a vehicle accident occurs.
Solution to Problem
(7) An illumination control device of the present disclosure includes: processing circuitry

configured to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected; and change a lamp color of the lamp body when an accident is detected; command an imaging device that performs imaging using visible light to image a vehicle interior when the accident is detected; specify a position of a site of a living body in the vehicle interior using an imaged image acquired in response to the command; and estimate the position of the site of the living body in the vehicle interior and command the lamp body to adjust light distribution with respect to the estimated position of the site of the living body when the position of the site of the living body in the vehicle interior is not specified.

Advantageous Effects of Invention

(8) According to the present disclosure, it is possible to provide an illumination control device that controls illumination in such a manner that biometric information can be obtained without depending on imaging conditions when a vehicle accident occurs.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a block diagram illustrating a configuration of an illumination control device according to a first embodiment and a peripheral device related thereto.

(2) FIG. **2** is a diagram illustrating an example of a correspondence relationship between a type of biometric information and a lamp color.

(3) FIG. **3** is a diagram illustrating an example of a correspondence relationship between a type of biometric information and a site of a living body.

(4) FIG. **4** is a diagram illustrating a first example of a hardware configuration of an illumination control device according to the present disclosure.

(5) FIG. **5** is a diagram illustrating a second example of the hardware configuration of the illumination control device according to the present disclosure.

(6) FIG. **6** is a flowchart illustrating a main process of the illumination control device and a main process of a biometric information acquiring device in the present disclosure.

(7) FIG. **7** is a flowchart illustrating an example of an illumination control process illustrated in FIG. **6**.

(8) FIG. **8** is a flowchart illustrating an example of a lamp color changing process illustrated in FIG. **7**.

(9) FIG. **9** is a flowchart illustrating an example of a biometric information acquiring process illustrated in FIG. **6**.

(10) FIG. **10** is a block diagram illustrating a configuration of an illumination control device according to a second embodiment and a peripheral device related thereto.

(11) FIG. **11** is a flowchart illustrating an example of an illumination control process according to the second embodiment.

(12) FIG. **12** is a flowchart illustrating an example of a light distribution adjusting process illustrated in FIG. **11**.

(13) FIG. **13** is a block diagram illustrating a configuration of an illumination control device according to a third embodiment and a peripheral device related thereto.

(14) FIG. **14** is a flowchart illustrating an example of a lamp color changing process according to the third embodiment.

DESCRIPTION OF EMBODIMENTS

(15) Hereinafter, in order to describe the present disclosure in more detail, embodiments for carrying out the present disclosure will be described with reference to the attached drawings.

First Embodiment

(16) FIG. **1** is a block diagram illustrating a configuration of an illumination control device **100** according to a first embodiment and a peripheral device related thereto.
(17) The illumination control device **100** controls illumination of a vehicle interior by commanding a lamp body **410** that illuminates the vehicle interior to turn on the lamp body **410** and designating a lamp color. In particular, the illumination control device **100** of the present disclosure issues a command to change a lamp color of the lamp body **410** when an accident of a vehicle occurs.
(18) The illumination control device **100** illustrated in FIG. **1** is communicably connected to a biometric information acquiring device **200**, an information source device **300**, and an output device **400**.
(19) An example of a detailed configuration of the illumination control device **100** will be described later.
(20) The biometric information acquiring device **200** acquires biometric information using an imaged image that is an image obtained by imaging a vehicle interior, and outputs the acquired biometric information.
(21) The biometric information in the present disclosure is biometric information that can be acquired using an imaged image, and is, for example, a pulse, a complexion, a bleeding state, respiration, or the like.
(22) The biometric information acquiring device **200** illustrated in FIG. **1** is communicably connected to the illumination control device **100**, the information source device **300**, and a communication device **500**.
(23) An example of a detailed configuration of the biometric information acquiring device **200** will be described later.
(24) The information source device **300** is a device that provides information used for a process by the illumination control device **100** or the biometric information acquiring device **200**.
(25) Specifically, the information source device **300** includes a vehicle sensor **310** and an imaging device **320** mounted on a vehicle.
(26) The vehicle sensor **310** outputs a signal related to vehicle information.
(27) The vehicle information in the present disclosure is information that makes it possible to determine an accident of a vehicle, a state related to the vehicle, a situation of a vehicle interior, and the like.
(28) The vehicle information is, for example, a vehicle speed, an acceleration, vehicle position information, seat sensor information, light sensor information, parking brake information, vehicle device failure notification information, or airbag operation notification information.
(29) The vehicle sensor **310** is constituted by a plurality of sensors that outputs a signal related to the vehicle information.
(30) Note that the vehicle sensor **310** may transmit a signal related to the vehicle information via an electronic control unit (ECU) (not illustrated) without directly transmitting a signal related to the vehicle information. The ECU is a control unit that controls each main operation of a vehicle. The ECU is connected to the illumination control device **100** by a wire harness (not illustrated), and can freely communicate with the illumination control device **100** by a communication method based on a controller area network (CAN) standard.
(31) The imaging device **320** is disposed at a position where a vehicle interior can be imaged, and images the vehicle interior.
(32) The imaging device **320** includes a first imaging element that converts visible light into an electrical signal, and outputs a visible light imaged image that is an image obtained by imaging the vehicle interior by the first imaging element receiving the visible light. The visible light imaged image is hereinafter simply referred to as an “imaged image”.
(33) The imaging device **320** in the present disclosure images the vehicle interior in response to a command from an imaging control unit **210**.
(34) In the first embodiment, specifically, when an accident of the vehicle occurs, the imaging

device **320** receives a command from the imaging control unit **210** of the biometric information acquiring device **200** and images the vehicle interior.

(35) Here, for example, a camera included in a driver monitoring system (DMS) may be used as the imaging device **320**.

(36) In this case, the imaging device **320** images occupants including a driver in the vehicle interior.

(37) When the camera included in the DMS is used, the imaging device **320** may include a second imaging element that converts infrared light into an electrical signal, and may have a function of outputting an infrared imaged image that is an image obtained by imaging the vehicle interior by the second imaging element receiving the infrared light. The infrared imaged image is hereinafter referred to as an “infrared image” in order to be distinguished from an imaged image that is a visible light imaged image.

(38) In this case, the imaging device **320** images infrared light and outputs an infrared image in a state where no vehicle accident occurs, and switches to visible light imaging by receiving a command from the imaging control unit **210**, and outputs an imaged image when a vehicle accident occurs.

(39) Note that the imaging device **320** may transmit an image via an electronic control unit (ECU) (not illustrated) without directly transmitting an image (imaged image or infrared image). The ECU is a control unit that controls each main operation of a vehicle. The ECU is connected to the biometric information acquiring device **200** or the illumination control device **100** by a wire harness (not illustrated), and can freely communicate with the biometric information acquiring device **200** or the illumination control device **100** by a communication method based on a controller area network (CAN) standard.

(40) The output device **400** is a control target device to which a command is issued from the illumination control device **100** of the present disclosure, and is specifically the lamp body **410**.

(41) The lamp body **410** is included in, for example, a vehicle interior illumination device, and illuminates the vehicle interior using visible light.

(42) The lamp body **410** is, for example, a room lamp, a map lamp, a foot lamp, a side step lamp, a door lamp, or an ambient lamp.

(43) The room lamp is the lamp body **410** that illuminates the entire vehicle interior. and is disposed, for example, on a ceiling portion of the vehicle interior.

(44) The map lamp is, for example, the lamp body **410** disposed for a driver's seat or an assistant driver's seat.

(45) The foot lamp is the lamp body **410** disposed in such a way as to illuminate feet of an occupant.

(46) The side step lamp is the lamp body **410** incorporated in a side step.

(47) The door lamp is the lamp body **410** incorporated in a door.

(48) The ambient lamp is a decorative lamp body incorporated in a dashboard, a center console, an edge of a side door, or the like.

(49) The lamp body **410** in the present disclosure may include all of the lamp bodies or some of the lamp bodies.

(50) The lamp body **410** illustrated in FIG. **1** switches between a lamp-on state and a lamp-off state in response to a command issued from the illumination control device **100**.

(51) The lamp body **410** also changes a lamp color in response to a command issued from the illumination control device **100**.

(52) Specifically, the lamp body **410** includes, for example, a light source obtained by aggregating a plurality of light emitting diodes (LEDs).

(53) As for the plurality of LEDs in the lamp body **410**, for example, three LEDs emit light beams having different wavelengths (for example, three primary colors of light (R, G, and B) such as red, green, and blue), respectively, and a wavelength component or a ratio of an intensity of each of the

light beams can be changed in response to a command issued from the illumination control device **100**.
(54) When an LED is used for the lamp body **410**, for example, power consumption can be reduced and a life thereof is longer as compared with a case where a lamp (for example, a halogen lamp) other than the LED is used. In addition, fine illumination control is possible.
(55) Illumination control in the present disclosure includes control of a lamp color, control of brightness, and control of light distribution.
(56) The control of a lamp color is, for example, to cause various colors to be output by adjusting and combining brightness values of the colors of RGB.
(57) The control of light distribution includes control of a light irradiation range and control of light directivity.
(58) The control of light distribution can be implemented by, for example, a method for lighting only some of the plurality of LEDs in the lamp body **410** without lighting all of the LEDs.
(59) The communication device **500** communicates with the outside of the vehicle.
(60) The communication device **500** performs, for example, wireless communication with an external device that collects information from a plurality of vehicles.
(61) The communication device **500** illustrated in FIG. **1** is a mode in which vehicle information acquired by a vehicle information acquiring unit **111** from the vehicle sensor **310** and biometric information acquired by the biometric information acquiring device **200** are transmitted to a server (not illustrated) via a communication unit **510**.
(62) Details of the illumination control device **100** will be described.
(63) The illumination control device **100** includes an accident detection unit **110**, a lamp body control unit **120**, and a lamp color changing unit **130**.
(64) In addition to the above components, the illumination control device **100** includes a control unit (not illustrated) and a storage unit (not illustrated). The control unit (not illustrated) controls, for example, the entire device including the components. The storage unit (not illustrated) stores, for example, information used in processes performed by the components or information generated in the processes.
(65) The accident detection unit **110** detects an accident of a vehicle using a signal output from the vehicle sensor **310** mounted on the vehicle.
(66) A method for detecting an accident of the vehicle using a signal from the vehicle sensor **310** can be implemented by a method using a known technique on the basis of a vehicle speed, an acceleration, vehicle position information, seat sensor information, light sensor information, parking brake information, vehicle device failure notification information, or airbag operation notification information.
(67) When detecting an accident of the vehicle, the accident detection unit **110** outputs an accident occurrence signal that is a signal indicating that an accident has occurred.
(68) When detecting an accident of the vehicle, the accident detection unit **110** outputs vehicle information indicated by the signal output from the vehicle sensor **310**.
(69) An example of a detailed configuration of the accident detection unit **110** will be described later.
(70) When an accident is detected by the accident detection unit **110**, the lamp body control unit **120** commands the lamp body **410** that illuminates the vehicle interior using visible light to turn on the lamp body **410**.
(71) Specifically, when receiving an accident detection signal that is a signal indicating that occurrence of an accident has been detected from the accident detection unit **110**, the lamp body control unit **120** outputs a lamp-on command signal that is a signal for commanding the lamp body **410** to turn on the lamp body **410**.
(72) When an accident is detected by the accident detection unit **110**, the lamp color changing unit **130** changes a lamp color of the lamp body **410**.

(73) FIG. **2** is a diagram illustrating an example of a correspondence relationship between a type of biometric information and a lamp color.
(74) The lamp color changing unit **130** executes a process by referring to, for example, first data **600** as illustrated in FIG. **2**. The first data **600** is information indicating a correspondence relationship between a type of biometric information and a lamp color, and is stored in advance in the storage unit (not illustrated).
(75) The lamp color changing unit **130** selects a type of biometric information and determines a lamp color by referring to the first data **600**. The lamp color changing unit **130** changes the lamp color of the lamp body **410** to the determined lamp color.
(76) An example of a detailed configuration of the lamp color changing unit **130** will be described later.
(77) A specific example of a detailed configuration of the accident detection unit **110** will be described.
(78) The accident detection unit **110** includes the vehicle information acquiring unit **111** and an accident determination unit **112**.
(79) The vehicle information acquiring unit **111** acquires a signal output from the vehicle sensor **310**.
(80) The vehicle information acquiring unit **111** acquires vehicle information using the acquired signal, and outputs the vehicle information to the accident determination unit **112**. In addition, the vehicle information acquiring unit **111** outputs the vehicle information to a communication control unit **240**.
(81) The accident determination unit **112** determines occurrence of an accident using the vehicle information. When determining that an accident has occurred, the accident determination unit **112** outputs an accident detection signal that is a signal indicating that occurrence of an accident has been detected.
(82) The accident determination unit **112** may output the vehicle information used for the determination together with the accident detection signal.
(83) The accident determination unit **112** may determine a type of accident using the vehicle information. The accident determination unit **112** that performs this determination outputs accident type information indicating the determined type of accident.
(84) A specific example of a detailed configuration of the lamp color changing unit **130** will be described.
(85) The lamp color changing unit **130** includes a biometric information type selecting unit **131** and a lamp color adjusting unit **132**.
(86) When an accident is detected by the accident detection unit **110**, the biometric information type selecting unit **131** selects a type of biometric information to be acquired by referring to type information indicating a type of biometric information stored in advance.
(87) Specifically, for example, when receiving an accident detection signal from the accident detection unit **110**, the biometric information type selecting unit **131** sequentially selects a type of biometric information to be acquired by referring to the type information.
(88) Alternatively, for example, when receiving an accident detection signal from the accident detection unit **110**, the biometric information type selecting unit **131** selects all types of biometric information to be acquired by referring to the type information.
(89) The biometric information type selecting unit **131** transmits information indicating the selected type to the lamp color adjusting unit **132**. In addition, the biometric information type selecting unit **131** transmits information indicating the selected type to a living body site specifying unit **220** described later.
(90) The lamp color adjusting unit **132** adjusts the lamp color of the lamp body **410** depending on the type of biometric information selected by the biometric information type selecting unit **131**.
(91) For example, the lamp color adjusting unit **132** determines the lamp color of the lamp body

410 depending on the type of biometric information by referring to the first data **600** illustrated in FIG. **2**, and transmits a command signal for commanding the lamp body **410** to turn on the lamp body **410** with the determined lamp color.
(92) For example, when the type of biometric information selected by the biometric information type selecting unit **131** is a pulse, the lamp color adjusting unit **132** performs adjustment in such a manner that a wavelength of the lamp color is equal to or more than 490 nm and equal to or less than 570 nm. As a result, the lamp color is so-called green, and the lamp body **410** irradiates a living body with green light.
(93) In addition, for example. when the type of biometric information selected by the biometric information type selecting unit **131** is a complexion, the lamp color adjusting unit **132** performs adjustment in such a manner that the lamp color is white. For example, the lamp body **410** is commanded to perform so-called all lamps-on. As a result, the lamp color is so-called white, and the lamp body **410** irradiates a living body with white light.
(94) A specific example of a detailed configuration of the biometric information acquiring device **200** will be described.
(95) The biometric information acquiring device **200** illustrated in FIG. **1** includes the imaging control unit **210**, the living body site specifying unit **220**, a biometric information acquiring unit **230**, and the communication control unit **240**.
(96) In addition to the above components, the biometric information acquiring device **200** includes a control unit (not illustrated) and a storage unit (not illustrated). The control unit (not illustrated) controls, for example, the entire device including the components. The storage unit (not illustrated) stores, for example, information used in processes performed by the components or information generated in the processes.
(97) When a vehicle accident occurs, the imaging control unit **210** commands the imaging device **320** to perform imaging.
(98) Specifically, when an accident is detected by the accident detection unit **110**. the imaging control unit **210** commands the imaging device **320** that performs imaging by receiving visible light to image the vehicle interior.
(99) Here, when the imaging device **320** can switch between imaging using visible light and imaging using infrared light, the imaging control unit **210** commands the imaging device **320** to perform imaging using infrared light in a state where no accident is detected by the accident detection unit **110**, and commands the imaging device **320** to perform imaging using visible light when an accident is detected by the accident detection unit **110**. That is, the imaging control unit **210** commands the imaging device **320** to perform imaging using infrared light until an accident is detected by the accident detection unit **110**.
(100) The living body site specifying unit **220** specifies the position of a site of a living body in the vehicle interior using an imaged image.
(101) Specifically, the living body site specifying unit **220** analyzes an imaged image acquired in response to a command of the imaging control unit **210**, and specifies the position of the site of the living body in the vehicle interior.
(102) In addition, the living body site specifying unit **220** extracts and outputs an image in a format capable of specifying the site of the living body. Specifically, for example, the living body site specifying unit **220** outputs an imaged image of the specified site together with information indicating the specified site.
(103) FIG. **3** is a diagram illustrating an example of a correspondence relationship between a type of biometric information and a site of a living body.
(104) The living body site specifying unit **220** includes, for example, second data **700** as illustrated in FIG. **3**. The second data **700** is information indicating a correspondence relationship between a type of biometric information and a site of a living body.
(105) The living body site specifying unit **220** extracts and outputs an imaged image of a site of a

living body corresponding to the type of biometric information selected by the biometric information type selecting unit **131** in the lamp color changing unit **130** by referring to the second data **700**.
(106) The site of the living body specified by the living body site specifying unit **220** is, for example, a head, a neck, a chest, an abdomen, a shoulder, or an arm, but is not limited to the above sites as long as the site can be specified from an imaged image.
(107) Note that the head or neck is considered to be a site where damage is most likely to lead to death, and is a site where a ratio of exposure of a bare skin is high. The chest or abdomen is considered to be a site where damage is likely to lead to death.
(108) Note that, in specifying a site by the living body site specifying unit **220**, the positions of all sites corresponding to a type of biometric information may be specified, or the sites may be sequentially specified. In specifying a site of a living body, all predetermined sites may be collectively specified, or each type of biometric information selected by the biometric information type selecting unit **131** may be separately specified. For example, when biometric information of a head is acquired, only a site of the head may be specified, and after the biometric information of the head is acquired, only a site of a chest may be specified in order to acquire biometric information of the chest.
(109) Here, an example of a method for specifying a living body site in the living body site specifying unit **220** will be described.
(110) In specifying a body site of an occupant, first, the living body site specifying unit **220** detects an occupant for whom a site of a living body is to be specified. The occupant is detected using a predetermined rule or machine learning from an imaged image as in occupant skeleton detection or occupant face detection. Note that it is conceivable that the occupant takes a posture different from that during normal driving due to damage caused by an accident, and it is difficult to detect the occupant. Therefore, the occupant may be detected using vehicle information acquired from the vehicle sensor **310**, or a detection result of the occupant before occurrence of an accident may be used. For example, the determination can be made using a result of skeleton detection or face detection of an infrared light image imaged before occurrence of an accident, or by determining that an occupant is in a seat when seat sensor information is equal to or more than a threshold. In addition, occupant detection accuracy may be further enhanced by using occupant detection information during vehicle traveling, such as information indicating that a vehicle speed or an acceleration is equal to or more than a certain threshold or information indicating that a parking brake is off, obtained from parking brake information.
(111) The biometric information acquiring unit **230** acquires biometric information using an imaged image including a site of a living body specified by the living body site specifying unit **220**.
(112) The biometric information acquiring unit **230** receives an imaged image output from the living body site specifying unit **220**.
(113) The biometric information acquiring unit **230** analyzes the imaged image and acquires biometric information.
(114) The biometric information acquiring unit **230** outputs the biometric information.
(115) The biometric information is acquired as follows, for example. Bleeding or complexion: Estimated from whether the color or shape of a specified body site in an imaged image coincides with a predetermined model or from a machine learning result. Pulse: Acquired by capturing a change in brightness of a face surface considered to be caused by a blood flow of an occupant in an imaged image. Since hemoglobin contained in blood absorbs green light, accuracy of information is increased by adjusting a lamp color to a green wavelength at which a change in brightness can be easily captured. Respiration: Specified from movement of a specified body site (chest, shoulder, abdomen, or the like) in an imaged image.
(116) The communication control unit **240** causes the communication device **500** to transmit the biometric information acquired by the biometric information acquiring unit **230**.

(117) The communication control unit **240** receives the biometric information output from the biometric information acquiring unit **230**.
(118) The communication control unit **240** outputs a command signal that is a signal for commanding the communication device to transmit the biometric information together with the biometric information.
(119) When receiving vehicle information from the vehicle information acquiring unit **111**, the communication control unit **240** also causes the communication device to transmit the vehicle information.
(120) Here, an example of a hardware configuration that implements the present disclosure will be described.
(121) FIG. **4** is a diagram illustrating a first example of a hardware configuration of the illumination control device **100** according to the present disclosure and a peripheral device. In FIG. **4**, a processing circuit **1000** is connected to the information source device **300**, the output device **400**, and the communication device **500**. The processing circuit **1000** can transmit and receive signals and can transmit and receive information to and from the information source device **300**, the output device **400**, and the communication device **500**.
(122) FIG. **5** is a diagram illustrating a second example of the hardware configuration of the illumination control device **100** according to the present disclosure and the peripheral device. In FIG. **5**, a processor **1001** and a memory **1002** are connected to the information source device **300**, the output device **400**, and the communication device **500**. The processor **1001** can transmit and receive signals and can transmit and receive information to and from the information source device **300**, the output device **400**, and the communication device **500**.
(123) Functions in the illumination control device **100** or the biometric information acquiring device **200** are implemented by a processing circuit.
(124) Specifically, functions of the accident detection unit **110**, the vehicle information acquiring unit **111**, the accident determination unit **112**, the lamp body control unit **120**, the lamp color changing unit **130** or a lamp color changing unit **130**B described later in a third embodiment, the biometric information type selecting unit **131**, the lamp color adjusting unit **132**, an external light color specifying unit **133** described later in the third embodiment, an imaging control unit **140** described later in a second embodiment and the third embodiment, a living body site specifying unit **150** described later in the second embodiment and the third embodiment, the biometric information acquiring device **200** or a biometric information acquiring device **200**A described later in the second embodiment and the third embodiment, the imaging control unit **210**, the living body site specifying unit **220**, the biometric information acquiring unit **230**, and the communication control unit **240** are implemented by a processing circuit.
(125) That is, the illumination control device **100** or the biometric information acquiring device **200** includes a processing circuit for executing a series of processes (processes in FIG. **7** to **9**, **11**, **12**, or **14**) from step ST**110** to step ST**140** illustrated in FIG. **6**. The processing circuit may be dedicated hardware or a central processing unit (CPU) that executes a program stored in a memory.
(126) When the processing circuit is dedicated hardware illustrated in FIG. **4**, for example, a single circuit, a composite circuit, a programmed processor, a parallel programmed processor, an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or a combination thereof corresponds to the processing circuit **1000**.
(127) Functions of the accident detection unit **110**, the vehicle information acquiring unit **111**, the accident determination unit **112**, the lamp body control unit **120**, the lamp color changing unit **130** or the lamp color changing unit **130**B described later in the third embodiment, the biometric information type selecting unit **131**, the lamp color adjusting unit **132**, the external light color specifying unit **133** described later in the third embodiment, the imaging control unit **140** described later in the second embodiment and the third embodiment, the living body site specifying unit **150** described later in the second embodiment and the third embodiment, the biometric information

acquiring device **200** or the biometric information acquiring device **200**A described later in the second embodiment and the third embodiment, the imaging control unit **210**, the living body site specifying unit **220**, the biometric information acquiring unit **230**, and the communication control unit **240** may be implemented by separate processing circuits, or may be collectively implemented by one processing circuit.

(128) When the processing circuit is the processor **1001** illustrated in FIG. **5**, functions of the accident detection unit **110**, the vehicle information acquiring unit **111**, the accident determination unit **112**, the lamp body control unit **120**, the lamp color changing unit **130** or the lamp color changing unit **130**B described later in the third embodiment, the biometric information type selecting unit **131**, the lamp color adjusting unit **132**, the external light color specifying unit **133** described later in the third embodiment, the imaging control unit **140** described later in the second embodiment and the third embodiment, the living body site specifying unit **150** described later in the second embodiment and the third embodiment, the biometric information acquiring device **200** or the biometric information acquiring device **200**A described later in the second embodiment and the third embodiment, the imaging control unit **210**, the living body site specifying unit **220**, the biometric information acquiring unit **230**, and the communication control unit **240** are implemented by software, firmware, or a combination of software and firmware. Software or firmware is described as a program and stored in the memory **1002**.

(129) The processor **1001** reads and executes the program stored in the memory **1002**, and thereby implements the functions of the units. That is, the illumination control device **100** includes the memory **1002** for storing a program that causes the above-described series of processes to be executed as a result when the program is executed by the processor **1001**. These programs cause a computer to execute a procedure or a method for implementing functions of the accident detection unit **110**, the vehicle information acquiring unit **111**, the accident determination unit **112**, the lamp body control unit **120**, the lamp color changing unit **130** or the lamp color changing unit **130**B described later in the third embodiment, the biometric information type selecting unit **131**, the lamp color adjusting unit **132**, the external light color specifying unit **133** described later in the third embodiment, the imaging control unit **140** described later in the second embodiment and the third embodiment, the living body site specifying unit **150** described later in the second embodiment and the third embodiment, the biometric information acquiring device **200** or the biometric information acquiring device **200**A described later in the second embodiment and the third embodiment, the imaging control unit **210**, the living body site specifying unit **220**, the biometric information acquiring unit **230**, and the communication control unit **240**.

(130) To the memory **1002**, for example, a nonvolatile or volatile semiconductor memory such as a random access memory (RAM), a read only memory (ROM), a flash memory, an erasable programmable read only memory (EPROM), or an electrically-EPROM (EEPROM), a magnetic disk, a flexible disk, an optical disc, a compact disc, a mini disc, a DVD, or the like corresponds.

(131) Some of the functions of the accident detection unit **110**, the vehicle information acquiring unit **111**, the accident determination unit **112**, the lamp body control unit **120**, the lamp color changing unit **130** or the lamp color changing unit **130**B described later in the third embodiment, the biometric information type selecting unit **131**, the lamp color adjusting unit **132**, the external light color specifying unit **133** described later in the third embodiment, an imaging control unit **140** described later in the second embodiment and the third embodiment, the living body site specifying unit **150** described later in the second embodiment and the third embodiment, the biometric information acquiring device **200** or the biometric information acquiring device **200**A described later in the second embodiment and the third embodiment, the imaging control unit **210**, the living body site specifying unit **220**, the biometric information acquiring unit **230**, and the communication control unit **240** may be implemented by dedicated hardware, and some thereof may be implemented by software or firmware. For example, the function of the vehicle information acquiring unit **111** may be implemented by the processing circuit **1000** as dedicated hardware, and

the function of the accident detection unit **110** may be implemented by the processor **1001** reading and executing a program stored in the memory **1002**.
(132) As described above, the processing circuit can implement each of the above functions by hardware, software, firmware, or a combination thereof.
(133) Processes according to the present disclosure will be described.
(134) FIG. **6** is a flowchart illustrating a main process of the illumination control device **100** and a main process of the biometric information acquiring device **200** in the present disclosure.
(135) FIG. **7** is a flowchart illustrating an example of an illumination control process illustrated in FIG. **6**.
(136) FIG. **8** is a flowchart illustrating an example of a lamp color changing process illustrated in FIG. **7**.
(137) FIG. **9** is a flowchart illustrating an example of a biometric information acquiring process illustrated in FIG. **6**.
(138) The illumination control device **100** starts a process, for example, at a timing when an engine is started in the vehicle ("Start" in FIG. **6**).
(139) The accident detection unit **110** waits until acquiring a signal output from the vehicle sensor **310**, and starts a process when acquiring the signal output from the vehicle sensor **310** (step ST**110**).
(140) The accident detection unit **110** determines occurrence of an accident using vehicle information indicated by the acquired signal, and detects the accident when determining that the accident has occurred.
(141) Specifically, the vehicle information acquiring unit **111** in the accident detection unit **110** acquires a signal output from the vehicle sensor **310**, and transmits vehicle information indicated by the acquired signal to the accident determination unit **112**.
(142) In addition, the vehicle information acquiring unit **111** outputs the vehicle information to the communication control unit **240**.
(143) The accident determination unit **112** in the accident detection unit **110** determines occurrence of an accident using the vehicle information.
(144) When determining that an accident has occurred, the accident determination unit **112** outputs an accident detection signal to each of the lamp body control unit **120**, the biometric information type selecting unit **131**, and the imaging control unit **210**.
(145) In a state where no accident is detected (step ST**120** "NO"), the accident detection unit **110** repeats the processes from step ST**110**.
(146) When the accident detection unit **110** detects an accident (step ST**120** "YES"), an illumination control process (step ST**130**) and a biometric information acquiring process (step ST**140**) are executed.
(147) In the configuration illustrated in FIG. **1**, the illumination control device **100** executes the illumination control process (step ST**130**), and the biometric information acquiring device **200** executes the biometric information acquiring process (step ST**140**).
(148) When the process in step ST**130** and the process in step ST**140** end, the process is ended ("End" in FIG. **6**).
(149) Details of the illumination control process (step ST**130**) will be described.
(150) In the illumination control device **100**, when the accident detection unit **110** outputs an accident detection signal, an illumination control process is started (Start illumination control process in FIG. **7**).
(151) The lamp body control unit **120** waits until receiving the accident detection signal output from the accident detection unit **110**.
(152) When receiving the accident detection signal output from the accident detection unit **110**, the lamp body control unit **120** commands the lamp body **410** to turn on the lamp body **410** (step ST**131**).

(153) The lamp color changing unit **130** waits until receiving the accident detection signal output from the accident detection unit **110**.
(154) When receiving the accident detection signal output from the accident detection unit **110**, the lamp color changing unit **130** proceeds to a lamp color changing process (step ST**132**).
(155) When the lamp color changing unit **130** ends the lamp color changing process (step ST**132**), the control unit (not illustrated) in the illumination control device **100** determines whether to end the illumination control process (step ST**133**).
(156) For example, when receiving a command to stop the process from the outside. the control unit determines to end the illumination control process.
(157) When determining not to end the illumination control process (step ST**133** “NO”), the illumination control device **100** repeats the lamp color changing process (step ST**132**).
(158) When determining to end the illumination control process (step ST**133** “YES”), the illumination control device **100** ends the illumination control process (End illumination control process in FIG. **7**).
(159) Details of the lamp color changing process (step ST**132**) will be described.
(160) The lamp color changing unit **130** starts the lamp color changing process (“Start lamp color changing process” in FIG. **8**).
(161) The biometric information type selecting unit **131** in the lamp color changing unit **130** selects a type of biometric information to be acquired (step ST**132-1**).
(162) The lamp color adjusting unit **132** in the lamp color changing unit **130** adjusts a lamp color of the lamp body **410** depending on the type of biometric information (step ST**132-2**).
(163) The control unit (not illustrated) determines whether all types of biometric information to be acquired have been selected (step ST**132-3**).
(164) When the control unit (not illustrated) determines that not all types of biometric information to be acquired have been selected (step ST**132-3** “NO”), the control unit performs control to repeat the processes from step ST**132-1**.
(165) When the control unit (not illustrated) determines that all types of biometric information to be acquired have been selected (step ST**132-3** “YES”), the control unit controls to end the lamp color changing process (“End lamp color changing process” in FIG. **8**).
(166) Details of the biometric information acquiring process (step ST**140**) will be described.
(167) The biometric information acquiring device **200** starts the biometric information acquiring process (“Start biometric information acquiring process” in FIG. **9**). The biometric information acquiring device **200** starts the biometric information acquiring process, for example, when receiving an accident detection signal from the accident determination unit **112** or when receiving an imaged image from the imaging device **320**.
(168) The living body site specifying unit **220** acquires an imaged image acquired by the imaging device **320** (step ST**141**).
(169) The living body site specifying unit **220** specifies the position of a site of a living body using the imaged image (step ST**142**).
(170) The living body site specifying unit **220** extracts and outputs an image in a format capable of specifying the site of the living body (step ST**143**).
(171) The living body site specifying unit **220** acquires a type of biometric information (step ST**144**).
(172) The biometric information acquiring unit **230** acquires biometric information using the type of biometric information and the image of the site of the living body (step ST**145**).
(173) The communication control unit **240** causes the communication device to transmit the biometric information acquired by the biometric information acquiring unit **230** (step ST**146**).
(174) The control unit (not illustrated) determines whether the biometric information acquiring process is ended (step ST**147**).
(175) When determining that the biometric information acquiring process is not ended (step ST**147**

"NO"), the control unit performs control to repeat the processes from step ST**144**.
(176) When determining that the biometric information acquiring process is ended (step ST**147** "YES"), the control unit controls to end the biometric information acquiring process ("End biometric information acquiring process" in FIG. **9**).
(177) Here, one modification of the first embodiment will be described.
(178) The present disclosure does not exclude that the illumination control device **100** and the biometric information acquiring device **200** are integrally constituted.
(179) The illumination control device **100** may include some or all of the components of the biometric information acquiring device **200**.
(180) In addition, the biometric information acquiring device **200** may include some or all of the components of the illumination control device **100**.
(181) As described above, the illumination control device according to the present disclosure includes: an accident detection unit to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; a lamp body control unit to command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected by the accident detection unit; and a lamp color changing unit to change a lamp color of the lamp body when an accident is detected by the accident detection unit.
(182) As a result, it is possible to provide an illumination control device that controls illumination in such a manner that biometric information can be obtained without depending on imaging conditions when a vehicle accident occurs.
(183) In the illumination control device according to the present disclosure, the lamp color changing unit includes: a biometric information type selecting unit to select a type of biometric information to be acquired by referring to type information indicating a type of biometric information stored in advance when an accident is detected by the accident detection unit; and a lamp color adjusting unit to adjust a lamp color of the lamp body depending on the type of biometric information selected by the biometric information type selecting unit.
(184) As a result, it is possible to provide an illumination control device capable of adjusting the lamp color of the lamp body depending on the type of biometric information.
(185) In the illumination control device according to the present disclosure, when the type of biometric information selected by the biometric information type selecting unit is a pulse, the lamp color adjusting unit performs adjustment in such a manner that a wavelength of the lamp color is equal to or more than 490 nm and equal to or less than 570 nm.
(186) As a result, it is possible to provide an illumination control device capable of accurately acquiring a pulse.
(187) In the illumination control device according to the present disclosure, an imaging control unit commands an imaging device to perform imaging using infrared light until an accident is detected by the accident detection unit.
(188) As a result, it is possible to provide an illumination control device capable of imaging a vehicle interior using infrared light in a state where no accident occurs.
(189) As described above, the biometric information acquiring device according to the present disclosure includes: an accident detection unit to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; a lamp body control unit to command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected by the accident detection unit; a lamp color changing unit to change a lamp color of the lamp body when an accident is detected by the accident detection unit; an imaging control unit to command an imaging device that performs imaging using visible light to image the vehicle interior when an accident is detected by the accident detection unit; a living body site specifying unit to specify the position of a site of a living body in the vehicle interior using an imaged image acquired in response to a command of the imaging control unit; a biometric information acquiring unit to acquire biometric information using an imaged image including the site of the living body specified

by the living body site specifying unit; and a communication control unit to cause a communication device to transmit biometric information acquired by the biometric information acquiring unit.

(190) As a result, it is possible to provide a biometric information acquiring device capable of obtaining biometric information without depending on imaging conditions when a vehicle accident occurs.

(191) As described above, the illumination control method according to the present disclosure includes: an accident detection step of detecting, by an accident detection unit, an accident of a vehicle using a signal output from a sensor mounted on the vehicle; a lamp body control step of commanding, by a lamp body control unit, a lamp body that illuminates a vehicle interior using visible light to turn on the lamp body when an accident is detected by the accident detection unit; and a lamp color changing step of changing, by a lamp color changing unit, a lamp color of the lamp body when an accident is detected by the accident detection unit.

(192) As a result, an illumination control device that controls illumination in such a manner that biometric information can be obtained without depending on imaging conditions when a vehicle accident occurs can be implemented using a computer or the like.

Second Embodiment

(193) A second embodiment is a mode in which an illumination control device includes an imaging control unit **140** and a living body site specifying unit **150**, and further adjusts light distribution with respect to the position of a site of a living body.

(194) FIG. **10** is a block diagram illustrating a configuration of an illumination control device **100**A according to the second embodiment and a peripheral device related thereto.

(195) The illumination control device **100**A illustrated in FIG. **10** includes the imaging control unit **140** and the living body site specifying unit **150**. In addition, the illumination control device **100**A includes a light distribution adjusting unit **160**.

(196) The imaging control unit **140** has a configuration similar to that of the imaging control unit **210** illustrated in FIG. **1**, and detailed description thereof is omitted.

(197) In the second embodiment, specifically, when an accident of a vehicle occurs, the imaging device **320** receives a command from the imaging control unit **140** of the illumination control device **100**A and images a vehicle interior.

(198) In addition to the function of the living body site specifying unit **220** illustrated in FIG. **1**, when the position of a site of a living body in the vehicle interior cannot be specified, the living body site specifying unit **150** outputs a specification impossible signal, which is a signal indicating that the position of the site of the living body cannot be specified, to the light distribution adjusting unit **160**.

(199) In addition, when the position of the site of the living body in the vehicle interior cannot be specified, the living body site specifying unit **150** estimates the position of the site of the living body.

(200) In addition, the living body site specifying unit **150** adjusts light distribution in such a manner that brightness of the estimated position of the site of the living body falls within a range between a first threshold and a second threshold.

(201) In addition, when a biometric information acquiring unit **230** cannot acquire biometric information, the living body site specifying unit **150** changes the first threshold or the second threshold.

(202) When the living body site specifying unit **150** cannot specify the position of the site of the living body in the vehicle interior, the light distribution adjusting unit **160** estimates the position of the site of the living body in the vehicle interior and commands a lamp body **410** to adjust light distribution with respect to the estimated position of the site of the living body.

(203) Specifically, when receiving a specification impossible signal from the living body site specifying unit **150**, the light distribution adjusting unit **160** estimates the position of the site of the living body in the vehicle interior.

(204) The light distribution adjusting unit **160** outputs a command signal that commands the lamp body **410** to adjust light distribution with respect to the estimated position of the site of the living body.
(205) For example, the light distribution adjusting unit **160** adjusts light distribution of visible light output from the lamp body **410** in such a manner that brightness of a site of a place where a body site is considered to be present is within a predetermined range by a predetermined method such as occupant skeleton detection.
(206) The adjustment of light distribution performed by the light distribution adjusting unit **160** includes at least one of the following three. (1) Selection of lamp body **410**: A lamp body to be turned on or off is selected from among a plurality of lamp bodies that illuminates a vehicle interior. (2) Selection of light source in lamp body: A light source to be lighted is selected from among a plurality of light sources (for example, LEDs) in a lamp body. (3) Light control of light source: Brightness of a light source is controlled.
(207) The adjustment of light distribution is determined in advance on the basis of. for example, a light control simulation result by a combination of the above (1), (2), and (3) performed at the time of optical design.
(208) The above (1), (2), and (3) are determined by referring to the simulation result depending on position information of a place where the body site is present or is estimated to be present, and brightness thereof.
(209) An example of adjustment of light distribution will be described.
(210) For example, when the brightness of the site specified by the living body site specifying unit **150** is equal to or more than a threshold and is too bright, the light distribution adjusting unit **160** controls light in such a manner that light distribution corresponding to the site decreases depending on a determination result, or increases light distribution behind the site. As a result, the brightness of the site is set to fall within a range between the first threshold and the second threshold.
(211) In addition, for example, when the brightness of the site specified by the living body site specifying unit **150** is equal to or less than the threshold and is too dark, the light distribution adjusting unit **160** controls light in such a manner that light distribution corresponding to the site increases depending on the determination result.
(212) Details of an illumination control process will be described.
(213) FIG. **11** is a flowchart illustrating an example of the illumination control process according to the second embodiment.
(214) In the illumination control device **100**A, when an accident detection unit **110** outputs an accident detection signal, the illumination control process is started (“Start illumination control process” in FIG. **11**).
(215) The lamp body control unit **120** waits until receiving the accident detection signal output from the accident detection unit **110**.
(216) When receiving the accident detection signal output from the accident detection unit **110**, the lamp body control unit **120** commands the lamp body **410** to turn on the lamp body **410** (step ST**231**).
(217) The lamp color changing unit **130** waits until receiving the accident detection signal output from the accident detection unit **110**.
(218) When receiving the accident detection signal output from the accident detection unit **110**, the lamp color changing unit **130** proceeds to a lamp color changing process (step ST**232**) or a light distribution adjusting process (step ST**233**).
(219) When the lamp color changing unit **130** ends the lamp color changing process (step ST**232**) or the light distribution adjusting process (step ST**233**), a control unit (not illustrated) in the illumination control device **100**A determines whether to end the illumination control process (step ST**234**).
(220) For example, when receiving a command to stop the process from the outside, the control

unit determines to end the illumination control process.
(221) When determining not to end the illumination control process (step ST**234** "NO"), the illumination control device **100**A repeats the lamp color changing process (step ST**232**) or the light distribution adjusting process (step ST**233**).
(222) When determining to end the illumination control process (step ST**234** "YES"), the illumination control device **100**A ends the illumination control process ("End illumination control process" in FIG. **11**).
(223) Details of the lamp color changing process (step ST**232**) are similar to those in FIG. **8**, and thus description thereof is omitted.
(224) Details of the light distribution adjusting process (step ST**233**) will be described.
(225) FIG. **12** is a flowchart illustrating an example of the light distribution adjusting process illustrated in FIG. **11**.
(226) The illumination control device **100**A starts the light distribution adjusting process ("Start light distribution adjusting process" in FIG. **12**).
(227) The living body site specifying unit **150** acquires an image imaged by the imaging device **320** (step ST**233-1**).
(228) The living body site specifying unit **150** specifies the position of each site of a living body using the acquired image (step ST**233-2**).
(229) The living body site specifying unit **150** determines whether the position of the site of the living body has been specified (step ST**233-3**).
(230) When determining that the position of the site of the living body cannot be specified (step ST**233-3** "NO"), the living body site specifying unit **150** estimates the position of the site of the living body (step ST**233-4**).
(231) When step ST**233-4** is executed, the process proceeds to step ST**233-5**.
(232) In step ST**233-5**, the light distribution adjusting unit **160** adjusts light distribution in such a manner that brightness of the estimated position falls within a range between the first threshold and the second threshold.
(233) When determining that the position of the site of the living body has been specified (step ST**233-3** "YES"), the living body site specifying unit **150** outputs an image in a format capable of specifying the site of the living body to the biometric information acquiring unit **230** (step ST**233-6**).
(234) The living body site specifying unit **150** determines whether biometric information has been acquired by the biometric information acquiring unit **230** (step ST**233-7**).
(235) When the biometric information acquiring unit **230** determines that the biometric information cannot be acquired (step ST**233-7** "NO"), the living body site specifying unit **150** changes the first threshold or the second threshold and proceeds to step ST**233-5**.
(236) After step ST**233-5** is executed, the processes are repeated from step ST**233-1**.
(237) When the biometric information acquiring unit **230** determines that biometric information has been acquired (step ST**233-7** "YES"), the light distribution adjusting process is ended ("End light distribution adjusting process" in FIG. **12**). Here, one modification of the second embodiment will be described.
(238) The present disclosure does not exclude that the illumination control device **100**A and the biometric information acquiring device **200**A are integrally constituted.
(239) The illumination control device **100**A may include some or all of the components of the biometric information acquiring device **200**A.
(240) In addition, the biometric information acquiring device **200**A may include some or all of the components of the illumination control device **100**A.
(241) The illumination control device according to the present disclosure further includes: an imaging control unit to command an imaging device that performs imaging using visible light to image a vehicle interior when an accident is detected by an accident detection unit; a living body

site specifying unit to specify the position of a site of a living body in the vehicle interior using an imaged image acquired in response to a command of the imaging control unit; and a light distribution adjusting unit to estimate the position of the site of the living body in the vehicle interior and commands a lamp body to adjust light distribution with respect to the estimated position of the site of the living body when the living body site specifying unit cannot specify the position of the site of the living body in the vehicle interior.

(242) As a result, it is possible to provide an illumination control device that controls illumination in such a manner that biometric information can be obtained more accurately without depending on imaging conditions when a vehicle accident occurs.

(243) As described above, the biometric information acquiring device according to the present disclosure further includes the light distribution adjusting unit to estimate the position of a site of a living body in a vehicle interior and commands a lamp body to adjust light distribution with respect to the estimated position of the site of the living body when the living body site specifying unit **220** cannot specify the position of the site of the living body in the vehicle interior.

(244) As a result, it is possible to provide a biometric information acquiring device capable of obtaining biometric information more accurately without depending on imaging conditions when a vehicle accident occurs.

Third Embodiment

(245) A third embodiment is a mode in which illumination with an influence of external light removed is performed.

(246) FIG. **13** is a block diagram illustrating a configuration of an illumination control device **100**B according to the third embodiment and a peripheral device related thereto.

(247) The illumination control device **100**B illustrated in FIG. **13** is the illumination control device **100**A illustrated in FIG. **10**, in which the lamp color changing unit **130** further includes an external light color specifying unit **133**.

(248) Among components similar to those of the illumination control device **100**A illustrated in FIG. **10**, description of the components described above is omitted.

(249) The external light color specifying unit **133** specifies a color of external light incident on a vehicle interior using an imaged image acquired by the imaging device **320** and lamp color information indicating a lamp color of a lamp body **410**A.

(250) When a type of biometric information selected by the biometric information type selecting unit **131** is a bleeding state or a complexion, the lamp color adjusting unit **132** performs adjustment in such a manner that a lamp color of the lamp body **410**A is a color obtained by subtracting the color of external light from the color indicated by the lamp color information.

(251) FIG. **14** is a flowchart illustrating an example of a lamp color changing process according to the third embodiment.

(252) The lamp color changing unit **130** starts the lamp color changing process (“Start lamp color changing process” in FIG. **14**).

(253) The biometric information type selecting unit **131** in the lamp color changing unit **130** selects a type of biometric information to be acquired (step ST**332-1**).

(254) The external light color specifying unit **133** acquires an imaged image and specifies the color of external light using the acquired image (step ST**332-2**).

(255) The lamp color of the lamp body **410** is determined depending on a type of biometric information, and adjustment is performed by subtracting the color of external light from the determined lamp color (step ST**332-3**).

(256) A control unit (not illustrated) determines whether all types of biometric information to be acquired have been selected (step ST**332-4**).

(257) When the control unit (not illustrated) determines that not all types of biometric information to be acquired have been selected (step ST**332-4** “NO”), the control unit performs control to repeat the processes from step ST**332-1**.

(258) When the control unit (not illustrated) determines that all types of biometric information to be acquired have been selected (step ST**332-4** “YES”), the control unit controls to end the lamp color changing process (“End lamp color changing process” in FIG. **14**).
(259) In the illumination control device according to the present disclosure, the lamp color changing unit further includes an external light color specifying unit to specify a color of external light incident on a vehicle interior using an imaged image acquired by an imaging device and lamp color information indicating a lamp color of a lamp body, and the lamp color adjusting unit performs adjustment in such a manner that the lamp color of the lamp body is a color obtained by subtracting the color of the external light from the color indicated by the lamp color information when the type of biometric information selected by the biometric information type selecting unit is a bleeding state or a complexion.
(260) As a result, it is possible to further provide an illumination control device capable of easily imaging and illuminating a site of a living body.
(261) Note that the embodiments of the present disclosure can be freely combined with one another, any component in each of the embodiments can be modified, or any component in each of the embodiments can be omitted within the scope of the present disclosure.
INDUSTRIAL APPLICABILITY
(262) The illumination control device according to the present disclosure is suitable for use in an illumination control device for a vehicle or the like.
REFERENCE SIGNS LIST
(263) **100**, **100**A, **100**B: illumination control device, **110**: accident detection unit, **111**: vehicle information acquiring unit, **112**: accident determination unit, **120**: lamp body control unit, **130**, **130**B: lamp color changing unit, **131**: biometric information type selecting unit, **132**: lamp color adjusting unit, **133**: external light color specifying unit, **140**: imaging control unit, **150**: living body site specifying unit, **200**, **200**A: biometric information acquiring device, **210**: imaging control unit, **220**: living body site specifying unit, **230**: biometric information acquiring unit, **240**: communication control unit, **300**: information source device, **310**: vehicle sensor, **320**: imaging device, **400**, **400**A: output device, **410**, **410**A: lamp body, **500**: communication device, **510**: communication unit, **600**: first data, **700**: second data, **1000**: processing circuit, **1001**: processor, **1002**: memory

Claims

1. An illumination control device comprising: processing circuitry configured to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected; change a lamp color of the lamp body when the accident is detected; command an imaging device that performs imaging using visible light to image a vehicle interior when the accident is detected; specify a position of a site of a living body in the vehicle interior using an imaged image acquired in response to the command; and estimate the position of the site of the living body in the vehicle interior and command the lamp body to adjust light distribution with respect to the estimated position of the site of the living body when the position of the site of the living body in the vehicle interior is not specified.
2. The illumination control device according to claim 1, wherein the processing circuitry is further configured to select a type of biometric information to be acquired by referring to type information indicating types of biometric information stored in advance when the accident is detected; and adjust the lamp color of the lamp body depending on the type of biometric information having been selected.
3. The illumination control device according to claim 2, wherein when the type of biometric information selected is a pulse, the processing circuitry performs adjustment in such a manner that

a wavelength of the lamp color is equal to or more than 490 nm and equal to or less than 570 nm.
4. The illumination control device according to claim 2, wherein the processing circuitry is further configured to specify a color of external light incident on the vehicle interior using the imaged image acquired by the imaging device and lamp color information indicating a lamp color of the lamp body; and perform adjustment in such a manner that the lamp color of the lamp body turns into a color obtained by subtracting the color of the external light from the color indicated by the lamp color information when the type of biometric information having been selected is a bleeding state or a complexion.
5. The illumination control device according to claim 1, wherein the processing circuitry commands the imaging device to perform imaging using infrared light until an accident is detected.
6. An illumination control device comprising: processing circuitry configured to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected; and change a lamp color of the lamp body when the accident is detected; select a type of biometric information to be acquired by referring to type information indicating types of biometric information stored in advance when the accident is detected; adjust the lamp color of the lamp body depending on the type of biometric information having been selected; and specify a color of external light incident on the vehicle interior using the imaged image acquired by the imaging device and lamp color information indicating a lamp color of the lamp body; and perform adjustment in such a manner that the lamp color of the lamp body turns into a color obtained by subtracting the color of the external light from the color indicated by the lamp color information when the type of biometric information having been selected is a bleeding state or a complexion.
7. A biometric information acquiring device comprising: processing circuitry configured to detect an accident of a vehicle using a signal output from a sensor mounted on the vehicle; command a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected; change a lamp color of the lamp body when the accident is detected; command an imaging device that performs imaging using visible light to image the vehicle interior when the accident is detected; specify a position of a site of a living body in the vehicle interior using an imaged image acquired in response to the command; acquire biometric information using the imaged image including the specified site of the living body; cause a communication device to transmit the biometric information having been acquired; and estimate the position of the site of the living body in the vehicle interior and command the lamp body to adjust light distribution with respect to the estimated position of the site of the living body when the position of the site of the living body in the vehicle interior is not specified.
8. An illumination control method comprising: Detecting an accident of a vehicle using a signal output from a sensor mounted on the vehicle; Commanding a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected; changing a lamp color of the lamp body when the accident is detected; commanding an imaging device that performs imaging using visible light to image a vehicle interior when the accident is detected; specifying a position of a site of a living body in the vehicle interior using an imaged image acquired in response to the command; and estimating the position of the site of the living body in the vehicle interior and commanding the lamp body to adjust light distribution with respect to the estimated position of the site of the living body when the position of the site of the living body in the vehicle interior is not specified.
9. An illumination control method comprising: detecting an accident of a vehicle using a signal output from a sensor mounted on the vehicle; commanding a lamp body that illuminates a vehicle interior using visible light to turn on when an accident is detected; changing a lamp color of the lamp body when the accident is detected; selecting a type of biometric information to be acquired by referring to type information indicating types of biometric information stored in advance when the accident is detected; adjusting the lamp color of the lamp body depending on the type of

biometric information having been selected; specifying a color of external light incident on the vehicle interior using the imaged image acquired by the imaging device and lamp color information indicating a lamp color of the lamp body; and performing adjustment in such a manner that the lamp color of the lamp body turns into a color obtained by subtracting the color of the external light from the color indicated by the lamp color information when the type of biometric information having been selected is a bleeding state or a complexion.